Supporting Information: Factorised Cartesian Atomic Cluster Expansion for Low-Cost Machine-Learning Interatomic Potentials

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S1. Architecture development history (aspirin, rMD17 split 1, 60 epochs)

Development proceeded by hypothesis-driven changes, each tested on the identical split and budget. Versions v0–v4 predate the unit parameterisation (lr 0.005–0.01); single 60-epoch runs vary by 20–30% between seeds (Table S4), so differences below that level are within noise.

S2. Training-stability probe

Constant learning rate, Adam (AMSGrad), batch 5; first optimiser step with a non-finite loss or gradient.

Table S1: Architecture iterations. E (meV) / F (meV/Å) test MAE. MACE-small control: 10.6 / 19.5.

version	change (hypothesis)	E / F	outcome
v0	Cartesian moments, scalar messages	16.4 / 30.5	baseline
v1	+ bond-projected directional invariants, element-pair radial	14.1 / 27.3	better
v1	layer norm → affine (magnitude information)	17.9 / 33.7	rejected
v2	+ rank-preserving tensor transport ($C = 96$)	9.2 / 21.2	better
v3	+ gated equivariant products ($C = 96$)	6.8 / 16.7	best accuracy; edge cost $\propto C$
v4	factorised edges, $B = 16$, $K = 32$	10.2 / 20.5	1.7× MACE-small speed
v4	wider node layer $C = 128$	11.0 / 22.0	no gain: bottleneck is edge rank
v4	$K = 64$ / $B = 32$ / $K_d = 48$	10.5/22.6, 8.8/18.1, 8.6/18.0	radial basis and transport width matter
v4	single pass (strictly local)	20.3 / 42.5	rejected
v5	per-atom equivariant RMS normalisation	20.7 / 32.7	stable but less accurate
v7	unit parameterisation, lr 0.01 / 0.02 / 0.04	12.5/24.1, 9.4/19.4, 8.9/17.6	adopted (lr 0.04)
v7	flagship $B = 16$, $r_c = 4.5$ Å	8.45 / 18.3	adopted
v7	$C = 64$, $r_c = 4.5$ Å	11.6 / 21.5	fastest; less accurate on aspirin
v8	CP edge densities (FACE-CP)	7.2 / 16.4	accuracy option

Additional observations: (i) at initialisation, 4- and 5-body invariants are 10^{-6}–10^{-9} of the 2-body ones, so they are invisible after layer normalisation; (ii) the optimiser can exploit the scale invariance of layer normalisation by inflating moment magnitudes, which reweights body orders and eventually overflows the second-derivative computation; (iii) fixed equivariant saturation ($s = 4$) removed all non-finite steps on Ag_3 but doubled the aspirin error (19.5/40.0); (iv) weight decay 10^{-3} worsened ethanol accuracy (4.56/17.1 vs 3.08/13.9 for the same seed); (v) the EMA copy of the model must also receive buffers, otherwise calibrated scales are lost at evaluation.

Table S2: Stability probe. "none" = no non-finite step within the probe length.

molecule, lr, steps	configuration	first non-finite step
aspirin, 0.01, 400	calibrated fixed scales, no layer norm	2
	same, max body order 3 / 4	4 / 2
	same, no transport and no gated products	3
	same, one pass / one pass with body order ≤ 3	17 / none (no learning)
	same + unit parameterisation	33
	layer norm (v4)	none (F-RMSE 315 meV/Å at step 400)
	layer norm + unit parameterisation	none (276)
ethanol, 0.01, 1500	layer norm (v4)	85
	layer norm + unit parameterisation	none (F-RMSE 283 → 107)
	layer norm + unit + calibrated scales	24

S3. Kernel engineering (exact, output-identical changes)

Table S3: Effect of implementation changes on the flagship (speed ratio to MACE-small, interleaved timing).

change	speed vs MACE-small
v4 PyTorch baseline	1.85×
+ fused density kernel (forward 2.1×, backward 150 → 40 ms)	2.5–2.7×
+ fused node algebra (6.3× on the node layer)	3.2×
+ constant-map transport	3.65×
+ r_c 4.5 Å (27% fewer edges)	4.1×
+ node kernel on symmetric storage	4.8×
idle-GPU measurement, 1,344 atoms	5.7×

For FACE-CP, replacing per-rank slicing of the $(N, M, 4, C)$ edge tensor by a single GEMM against all 20 monomials raised the speed ratio from 2.0× to 2.9×.

Table S4: Seed variance and ensembles (aspirin unless stated; E meV / F meV/Å).

set	result
flagship, 400 epochs, 5 seeds	2.97–3.42 / 8.82–9.60
ethanol flagship, 60 epochs, 3 seeds	2.87–3.93 / 13.3–16.3
ethanol MACE-small, 60 epochs, 3 seeds	3.02–3.38 / 13.0–14.5
3 / 5 flagship ensemble	2.73 / 7.35; 2.73 / 7.09
7 / 11 flagship-class ensemble	2.64 / 6.84; 2.50 / 6.67
11-model teacher incl. distilled students	2.55 / 6.53
ethanol, 3 / 4 member ensemble	0.61 / 3.86; 0.60 / 3.73

Table S5: Distillation variants that did not help (aspirin, single models).

variant	E / F
flagship student, 1 teacher batch (19k)	3.49 / 8.32
flagship student, 2 batches (38k), same steps	3.55 / 8.55
CP student, 1 batch	3.07 / 7.83
CP student, 3 batches incl. 600 K (56k), same steps	3.37 / 8.32
rank-3 transport student	3.62 / 8.46
polynomial (linear per-pass) readout, 400 epochs	3.59 / 9.75
rank-3 transport, 400 epochs, seeds 0/1	2.91/8.30, 3.24/9.17
per-element product gates, 400 epochs	2.97 / 8.82 (within seed spread)

S4. Ensembles, seed variance and additional distillation results

S5. Error metrics, scaling and published reference values

Table S6 complements the MAE values of the main text with root-mean-square errors on the same 1000 test structures. RMSE/MAE ratios of 1.4–1.6 (energy) and 1.5–1.9 (force) are similar for FACE and MACE, so the ranking does not depend on the metric.

Table S6: Energy (meV) and force (meV/Å) MAE and RMSE, rMD17 test set (1000 structures).

model	E MAE	E RMSE	F MAE	F RMSE
aspirin				
flagship, 400 epochs	3.13	4.42	9.08	13.86
FACE-CP $C = 96$, 400 epochs	2.88	4.59	8.73	13.61
flagship, distilled	3.13	4.37	8.04	12.29
FACE-CP $C = 96$, generation 3	2.50	3.60	6.97	10.69
FACE-CP $C = 160$, generation 4	2.26	3.23	6.74	10.43
MACE-small, 60 epochs	10.60	15.13	19.51	28.52
MACE-medium, 60 epochs	9.13	12.85	17.80	26.33
ethanol				
FACE-CP, 400 epochs	0.78	1.21	4.84	8.52
FACE-CP, distilled	0.68	1.06	4.22	7.92
MACE-small, 400 epochs	1.11	1.73	6.76	12.02

Table S7: Inference speed relative to MACE-small versus system size (energy + forces, float32, interleaved timing rounds, median; periodic boxes of n^3 randomly oriented aspirin molecules on a cubic grid, $n = 2$–5).

atoms	168	567	1,344	2,625
FACE flagship	2.34×	4.67×	5.64×	5.43×
FACE-CP $C = 96$	2.37×	2.77×	3.20×	2.91×

Table S8: Training-step cost: forward pass, forces with `create_graph`, E/F loss, backward pass and AMSGrad step on five aspirin molecules (median of seven interleaved rounds of ten steps). Parameter counts are those of the benchmark construction.

model	parameters	step (ms)	relative to MACE-small
MACE-small	204,432	97.2	1.00
MACE-medium	604,432	130.9	1.35
FACE flagship	1.56 M	93.4	0.96
FACE-CP $C = 96$	1.31 M	92.6	0.95
FACE-CP $C = 160$	3.36 M	95.1	0.98

Table S9: Published rMD17 MAE values (E meV / F meV/Å, 950 training structures, converged training) for context; the 60-epoch results of the main text are not converged and are not directly comparable.

model	malonaldehyde	uracil	toluene	ethanol
NequIP	0.8 / 5.1	0.4 / 3.1	0.3 / 1.6	0.4 / 2.8
Allegro	0.6 / 3.6	0.6 / 1.8	0.4 / 1.8	0.4 / 2.1
MACE	0.8 / 4.1	0.5 / 2.1	0.5 / 1.5	0.4 / 2.1

Table S10: Cost of the physics operators relative to the local flagship (fused kernels and spline radial functions), energy and forces, periodic aspirin boxes; interleaved timing rounds, medians. Electrostatics: latent charges in coulomb mode with smooth particle-mesh Ewald; dispersion with a 10 Å pair list.

atoms	electrostatics	dispersion	ZBL	all three
567	+62%	+12%	+8%	+81%
1,344	+32%	+9%	+5%	+44%
2,625	+17%	+10%	+3%	+28%
4,536	+12%	+12%	+2%	+23%

Table S11: Inference time versus model width at 1,344 atoms, relative to MACE with 64 channels measured in the same interleaved rounds (energy and forces, float32).

model	time relative to MACE, 64 channels
MACE, 32 channels	0.577
MACE, 64 channels	1.000
MACE, 128 channels	1.874
FACE $C = 32$	0.163
FACE $C = 64$	0.168
FACE $C = 96$	0.181
FACE $C = 128$	0.195
FACE $C = 192$	0.229
FACE $C = 256$	0.271
FACE-CP $C = 64$	0.265
FACE-CP $C = 96$	0.342
FACE-CP $C = 160$	0.503

Table S12: Peak GPU memory (GB) for one energy and force evaluation, periodic aspirin boxes, float32. OOM: exceeds the 12 GB cap.

atoms	FACE flagship	FACE-CP $C = 96$	MACE, 64 ch.	MACE, 128 ch.
168	0.10	0.21	0.36	0.66
567	0.31	0.71	1.13	2.17
1,344	0.70	1.69	2.67	5.12
2,625	1.39	3.38	5.20	9.97
4,536	2.30	5.66	8.98	OOM

Table S13: Learning curve of the FACE flagship on rMD17 aspirin: nested training subsets, the same 50 validation and 1000 test structures; energy (meV) and force (meV/Å) MAE.

N	epochs	E	F	note
50	2000	24.60	50.30	
100	1000	10.74	32.20	
250	400	7.13	19.74	
500	400	4.18	12.77	stopped at epoch 380
950	400	3.13	9.08	

S6. Hyperparameters

Two message-passing steps; eight Bessel functions with polynomial envelope ($p = 6$); radial MLP width 64 with element-pair embedding; $K_d = 16$ directional and transport channels; nine invariant families with layer normalisation and an MLP of widths $(9C + C + 4) \to 2C \to 2C \to C + 1$; SiLU activations; unit parameterisation of all linear layers. Optimiser: AMSGrad/AdamW, weight decay 0, peak learning rate 0.04 (one-cycle, 5% warm-up, initial and final divisors 10 and 100), batch 5, gradient clipping 10, EMA 0.99, energy-weighted final 25% of epochs. Distillation: 25 epochs over DFT×4 plus teacher data; DFT-only fine-tune 60 epochs at 4×10^{-3} (half energy-weighted); energy-weighted fine-tune 60 epochs at 2×10^{-3} with $(w_E, w_F) = (1000, 10)$. Latent charges: Gaussian width 0.5 Å, Ewald splitting $\gamma' = r_c/3.5$, SPME grid spacing $\gamma'/4$, order-4 B-splines.

S7. Entries of the transport map

With the pooled sums $S_{(p,y)} = \bar{n}^{-1} \sum_j Y_p(\hat{\boldsymbol{u}}_{ij}) \phi_b(r_{ij}) T_{j,y}$ defined in the section on tensor transport of the main text (indices b and k omitted), $p \in \{0, 1, 2, 3\}$ for the monomials $\{1, u_1, u_2, u_3\}$ and y running over the components V_a and M_{ab} ($a \le b$) of the neighbour tensor, the 18 outputs of the (36×18) map are

$$\sum_j \phi_b \, V_{j,a} = S_{(0,V_a)}, \qquad \sum_j \phi_b \, (\boldsymbol{M}_j \hat{\boldsymbol{u}})_a = \sum_{c=1}^{3} S_{(c,M_{ac})},$$

$$\sum_j \phi_b \, M_{j,ab} = S_{(0,M_{ab})}, \qquad \sum_j \phi_b \, \mathrm{sym}(\hat{\boldsymbol{u}} \otimes \boldsymbol{V}_j)_{ab} = \tfrac{1}{2}\big(S_{(a,V_b)} + S_{(b,V_a)}\big),$$

for $a, b = 1, 2, 3$, with $M_{ac} = M_{ca}$ read from symmetric storage. All other entries are zero, and every non-zero entry is 1 or $\frac{1}{2}$. The rank-3 option extends p to the ten monomials of degree ≤ 2 and adds the outputs $\mathrm{sym}(\hat{\boldsymbol{u}} \otimes \boldsymbol{M}_j)$ and $\mathrm{sym}(\hat{\boldsymbol{u}} \otimes \hat{\boldsymbol{u}} \otimes \boldsymbol{V}_j)$, which are symmetrised sums of the same kind ((90×38) map). The maps are generated programmatically and tested against explicit per-edge evaluation.

S8. Learning curves

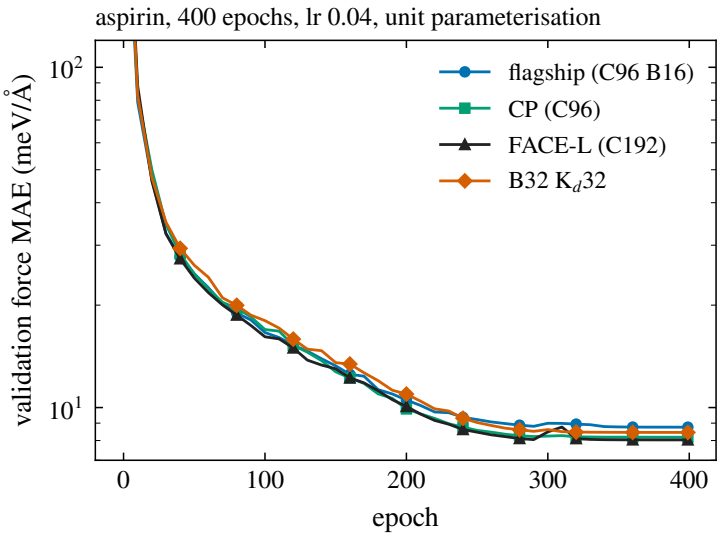


Figure S1: Validation force MAE during 400-epoch training of four configurations (aspirin).